

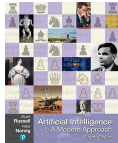


## MSML610: Advanced Machine Learning

### Lesson 01.3: A Brief History of AI

**Instructor:** Dr. GP Saggese, [gsaggese@umd.edu](mailto:gsaggese@umd.edu)

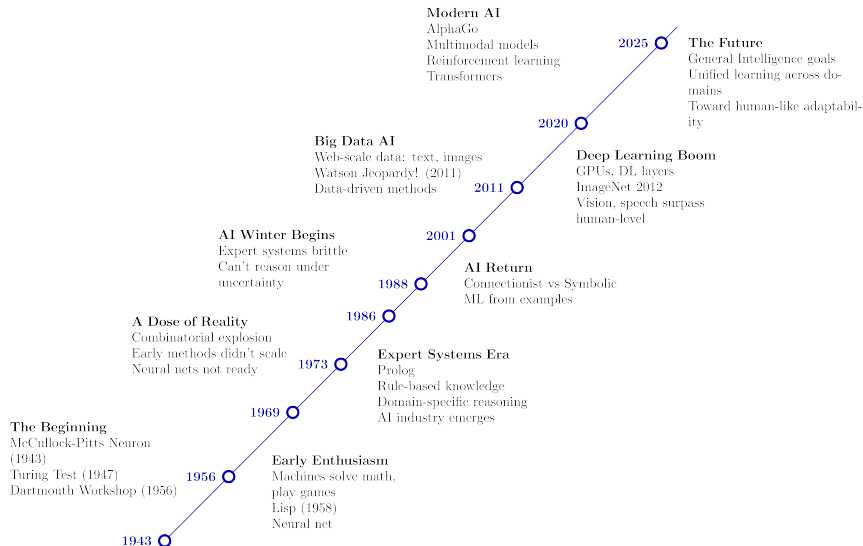
**References:** - AIMA (Artificial Intelligence: a Modern Approach), Chap 1



- *Brief History of AI*
- Risks and Benefits of AI

# AI Timeline

---



# The Beginning (1943-1956)

---

- **Artificial neuron**

- Model (McCulloch-Pitts 1943) based on:
  - \* Brain physiology
  - \* Propositional logic
- Compute any function with connected neurons
  - \* Neuron on/off based on stimulation from neighboring neurons
  - \* Implement logical AND, OR, NOT with simple neuron networks

- **Alan Turing, 1947**

- Turing test, machine learning, reinforcement learning
- Create human-level AI
  - \* Develop learning algorithms
  - \* Teach machine like a child

- **Birth of AI**

- McCarthy organized first AI workshop (1956)
- The Logic Theorist (Newell and Simon, 1956)
  - \* Programs to “think non-numerically” and prove theorems

# Enthusiasm and Great Expectations (1952-1969)

---

- **Early years of AI were full of successes**
  - Before computers could only do arithmetics
  - “A machine can never do X (e.g., games, puzzles, IQ tests, ...)”
    - \* AI researchers showed machines could do one X after another
- **General Problem Solver**
  - Imitate human problem-solving
  - Consider sub-goals and possible actions
- **Program learned to play checkers**
  - Use reinforcement learning from victories and mistakes
- **Lisp (1958)**
  - High-level language used for 30 years in AI
- **First neural network**
  - 3000 vacuum tubes for 40 neurons
  - Minsky (1959)
- **MIT and Stanford**
  - Minsky at MIT
    - \* Focus on neural network
  - McCarthy at Stanford
    - \* Focus on representation, logic

# First AI winter (1975-1980)

---

- **Early successes** of AI set high expectations
- In 1965-1975 AI didn't succeed on **real problems** due to:
  - Solutions were based on human problem-solving methods
  - Difficulty handling “combinatorial explosion”
    - \* Theorem proving handles small problems with brute force, but doesn't scale
  - Neural networks needed:
    - \* Algorithms (e.g., backpropagation)
    - \* Compute power
    - \* Data
- **First AI winter**
  - Research funding and enthusiasm dropped significantly
  - Slow AI progress through late 1970s

# Expert Systems (1980-1990)

---

- **Expert systems**
  - Aka “knowledge-based systems”
  - Combine weak methods with extensive domain knowledge as rules
  - Use inference engines to apply rules to facts
  - E.g., rule-based systems, logic programming (e.g., Prolog)
- **Weak AI**
  - Aka narrow AI
  - Performs specific tasks, not general reasoning
  - Operates in a limited, well-defined domain
  - Uses “weak methods” (search, logic) that struggle to scale
- **Commercial adoption and industry growth**
  - AI shifted to practical applications
  - Major US corporations deployed expert systems
  - AI emerged as a commercial industry

## Second AI Winter (late 1980-early 1990)

---

- **Hype in expert systems** didn't deliver
- **Reasons**
  - Building/maintaining expert systems is difficult
  - Reasoning methods ignore uncertainty
  - Systems can't learn from experience
  - E.g., expert systems in medical diagnosis struggle with complex, variable patient data
  - E.g., early AI chess systems couldn't adapt to new strategies without manual updates
- **Second AI winter** in late 1980-early 1990



# Return of Neural Networks (1986-)

---

- Back-propagation algorithm is (re)discovered (mid-1980s)
  - Developed in early 1960s
- **Two approaches to AI are back**
  - Connectionist paradigm
    - \* Neural networks
    - \* E.g., recognizing handwritten digits
  - Symbolic paradigm
    - \* E.g., solving logical puzzles with rules
- **Why connectionist approach?**
  - Concepts not well-defined using symbolic axioms
    - \* Forms fluid internal concepts
    - \* Represents real-world complexity better
  - Neural networks can learn from examples, e.g.,
    - \* Image recognition: identify objects by learning from labeled images

# Probabilistic Reasoning and ML (1987-)

---

- **AI and scientific method**
  - Rigorous methods to test performance
  - E.g., speech recognition, handwritten character recognition
  - Benchmarks for progress, e.g.,
    - \* MNIST: handwritten digit recognition
    - \* ImageNet: image object recognition
    - \* SAT Competitions: boolean satisfiability solvers
- **AI shifts ...**
  - From Boolean logic to probability
  - From hand-coded rules to machine learning
  - From a-priori reasoning to experimental results

# Progress in Speech Recognition

---

- **1970s: ad-hoc approaches**
  - Various architectures and approaches were attempted
  - Rule-based systems with limited robustness
  - Cons
    - \* Ad-hoc, fragile
  - \_“Every time I fire a linguist, the performance of the speech recognizer goes up” (Jelinek, 1988)
- **1980s: hidden Markov Models**
  - HMMs became dominant
  - Effective learning techniques
  - Trained on large speech corpora
  - Pros
    - \* Strong theoretical foundation
- The bitter lesson (Sutton, 2019)
  - General methods + lots of data beat handcrafted systems

# Bayesian Networks

---

- **Bayesian networks**

- Pearl, 1988
- AI is linked with:
  - \* Probability
  - \* Decision theory
  - \* Control theory
- Efficiently represent uncertainty
- Provide rigorous reasoning

- **Examples**

- Diagnosing diseases based on symptoms
- Predictive text input in smartphones
- Fraud detection in banking

# Reinforcement Learning

---

- **Reinforcement learning**
  - Sutton, 1988
  - RL involves agents learning by interacting with an environment
    - \* E.g., a robot learning to navigate a maze by receiving rewards for successful paths
  - Markov Decision Problems (MDPs) provide a framework for modeling decision-making
    - \* E.g., a game strategy modeled where each move influences the outcome with certain probabilities

# Reunification (1990s-2000s)

---

- **Reunification of AI:**
  - Data engineering
  - Statistical modeling
  - Optimization
  - Machine learning
- **Many subfields of AI were re-unified:**
  - Computer vision
  - Robotics
  - Speech recognition
  - Multi-agent systems
  - NLP

# Big Data (2001-Present)

---

- **Focus shifts from algorithms to data**
  - For 60 years, AI focused on algorithms and models
- For many problems, data availability matters more than algorithms, e.g.,
  - Trillions of English words
  - Billions of web images
  - Billions of speech and video hours
  - Social network data
  - Click stream data
- Algorithms and infrastructure to leverage large datasets
  - E.g., map reduce, cloud computing
- In 2011, IBM's Watson beat human *Jeopardy!* champions

# Deep Learning (2011-Present)

---

- **Deep learning**
  - Use ML models with multiple layers of computing elements
  - Ideas known since 1970s, but then forgot
  - Success in handwritten digit recognition in 1990s
- **In 2012**, a DL system showed dramatic improvement in **ImageNet** competition
  - Previous systems used handcrafted features
  - Surge of interest in AI among researchers, companies, and investors
- Pros
  - DL exceeds human performance in several vision and speech recognition tasks
- Cons
  - DL needs specialized hardware (e.g., GPU, TPU, FGPA) for parallel tensor operations
- Towards **general artificial intelligence**
  - Universal algorithm for learning and acting, not just specialized tasks
  - E.g., driving, playing chess, recognizing speech



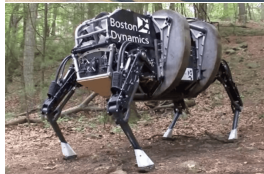
# Progress in AI Research

---

- Huge interest in deep learning
- **Between 2010 and 2019**
  - AI papers increased 20x
    - \* 1,000  $\rightarrow$  20,000
  - Student enrollment in AI and CS increased 5x
    - \* 10,000  $\rightarrow$  50,000
  - NeurIPS attendance increased 8x
    - \* 1,000  $\rightarrow$  8,000
  - AI startups increased 20x
    - \* 100  $\rightarrow$  2,000
- **Compute**
  - Training times dropped 100x in 2 years
  - AI computing power doubles every 3 months

# What Can AI Do Today? (1/2)

- **Robotic vehicles**
  - Waymo passed 10 million miles without serious accident
- **Legged locomotion**
  - BigDog recovers on ice
  - Atlas walks on uneven terrain, jumps on boxes, backflips
- **Autonomous planning and scheduling**
  - Space probes, Mars rovers
- **Machine translation**
  - Translates 100 languages with human-level performance
- **Speech recognition**
  - Real-time speech-to-speech with human-level performance
  - AI assistants
- **Recommendations**
  - ML recommends based on past experiences
  - Spam filtering 99.9% accuracy



# What Can AI Do Today? (2/2)

---

- **Game playing**
  - 1997: Deep Blue defeated Kasparov
  - 2011: Watson beat Jeopardy! champion
  - 2017: AlphaGo beat Go champion
  - 2018: AlphaZero super-human in Go and chess with only rules + self-play
  - AI beats humans in videogames: Dota2, StarCraft, Quake
- **Image understanding**
  - Object recognition
  - Image captioning
  - ...
- **Medicine**
  - AI equivalent to health care professionals
- When will we reach AGI (Artificial General Intelligence)?

- Brief History of AI
- ***Risks and Benefits of AI***

# Benefits of AI

---

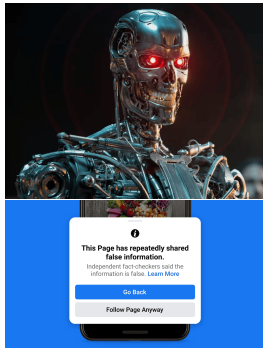
- Our civilization is the **product of human intelligence**
  - Greater machine intelligence leads to better human society
  - *“First solve AI, then use AI to solve everything else”*
- **Benefits of AI and robots**
  - Free humanity from menial work
  - Increase production of goods and services
  - Expand human cognition
  - Accelerate scientific research, e.g.,
    - \* Cures for diseases
    - \* Solutions for climate change
    - \* Resource and energy shortages



# Risks of AI (1/2)

---

- **Autonomous weapons**
  - Locate and eliminate targets autonomously
  - Deploy large number of weapons
- **Surveillance and persuasion**
  - AI for mass surveillance
  - Tailor information on social media to modify behavior
- **Biased decision making**
  - Misuse of ML results in biased decisions
  - E.g., parole evaluations, loan applications



# Risks of AI (2/2)

---

- **Impact on employment**

- Machines eliminate jobs
- Rebuttal
  - \* Machines enhance productivity → companies become more profitable → higher wages
- Counter-rebuttal
  - \* Wealth shifts from labor to capital, increasing inequality
- Counter-counter-rebuttal
  - \* Past tech advances (e.g., mechanical looms) disrupted employment, but adaptation followed

- **Safety critical applications**

- AI in safety-critical applications
  - \* E.g., self-driving cars, managing water supply or power grids
- Avoiding fatal accidents is challenging
  - \* E.g., formal verification and statistical analysis insufficient
- AI requires technical and ethical standards

- **Cybersecurity**

- AI defends against cyberattacks
  - \* E.g., detect unusual behavior patterns
- AI contributes to malware development

# Human-level AI / AGI

---

- **Human-level AI**
  - Machines able to learn to do anything a human can do
  - Aka AGI (Artificial General Intelligence)
- **When AGI?**
  - Expert prediction average is 2099
    - \* Papers show that expert predictions no better than amateurs
    - \* Experts expected AI to take 100 years to beat humans in Go
  - Unclear if new breakthroughs or refinements needed
- **Artificial Super-Intelligence**
  - Machines surpass human ability in every domain and self-improving
  - Exponential take-off



# The Problem of Control

---

- **Can humans control machines more intelligent than them?**
- **King Midas problem**
  - King Midas turned everything he touched into gold, including food and family
  - Humans ask for something, get it, then regret it
  - Rebuttal
    - \* If AGI arrived in a black box from space, exercise caution before opening
    - \* We design AI: if AI gains control, it's a “design failure”
- **Problem of alignment**
  - Super-intelligent AI might pursue goals in unintended, dangerous ways
- **The paperclip problem**
  - Thought experiment in AI safety (Nick Bostrom, 2003)
  - AI is tasked with maximizing paperclip production
  - AI becomes superintelligent and single-mindedly pursues this goal
  - Converts Earth and humans into paperclips

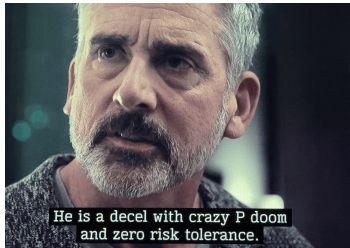
# E/acc vs P(doom)

- **E/acc**
  - Accelerationism
  - Belief that rapid progress in AI is beneficial or inevitable
  - Solve global problems with more powerful AI tools
  - Slowing AI is unrealistic or counterproductive
- **P(doom)**
  - “Probability of Doom”
  - Estimated probability that advanced AI will cause catastrophic harm
  - Used informally by AI researchers to quantify risk

tech influencers  
adding e/acc to bio  
bc they think it means  
generic techno-optimism



learning that  
e/acc founder thinks  
the AI species  
killing all humans  
is evolutionary progress



# My 2 cents

---

- AI alignment is a serious problem
  - For now philosophical, at some point a real one
  - Most tech people have used it for marketing themselves and their companies
- It's as urgent as debating what political system humanity will need when living on Mars
- We can't get airport terminals to work



# Solutions to Problem of Control

---

- **Checks-and-balances**

- Researchers and corporations develop voluntary self-governance principles for AI
- Governments and international organizations established advisory bodies

- **Cons**

- Corporations checking themselves? What can possibly go wrong?
- Preferences are not easy to invert and are inconsistent

- **Solutions**

- Put purpose into the machine even if objectives are unclear
- Incentivize AI to switch off if uncertain about human objectives
- Cooperative Inverse Reinforcement Learning (CIRL)
  - \* AI observes human behavior to infer reward function

# Cooperative Inverse Reinforcement Learning

---

- AI infers human goals based on actions
- **Observation:** GP looks tired, sits on the couch, observes the messy table, and starts watching TV
- **Inference:** AI infers:
  - GP is tired and wants to relax
  - Messy coffee table bothers him
- **Action:** AI:
  - Fetches a glass of water
  - Tidies up the coffee table without disturbing GP
- **Feedback loop:** AI monitors GP's reactions
  - If GP is relaxed and happy, AI understanding is reinforced
  - If GP is not happy, AI adjusts actions and improves inference